Real-time Object Classification for Embedded Systems Using Machine Learning

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Background

Machine learning models are powerful but often require expensive GPUs to run in real time. For smaller-scale applications, such hardware is often inaccessible.

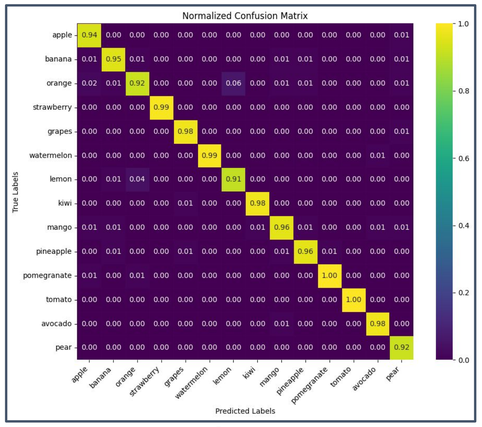
This project demonstrates that a low-cost Raspberry Pi system can detect and classify fruit in real time, even under challenging conditions such as variable lighting and cluttered backgrounds.

Research Questions

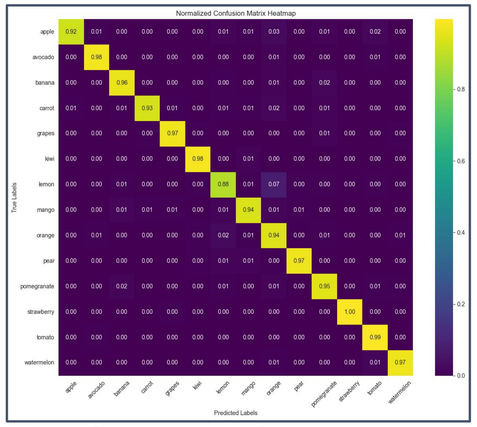
Can we build a low-cost, real-time fruit recognition system that runs fully on a Raspberry Pi while keeping accuracy high enough for practical use?

Methods and Materials

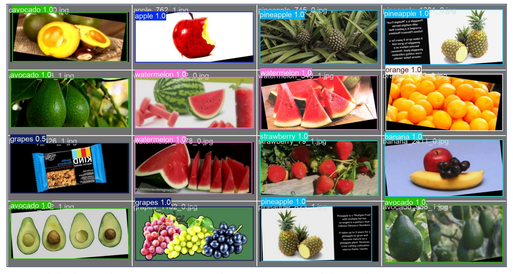
- **Hardware/Software:** Raspberry Pi 4B (4 GB) running Raspberry Pi OS Bullseye (64-bit), USB 1080p camera, Python with TensorFlow Lite, ONNX Runtime, Tkinter UI.
- **Dataset:** LVIS Fruits & Vegetables (Kaggle), cleaned to 14 classes, balanced to ~600 images each with rotation, brightness, shear, and zoom augmentation.
- **Model Pipeline:** YOLOv8 (ONNX) + MobileNetV2 (TFLite) integrated for detection + classification with four modes (YOLO-only, classifier-only, YOLO→classifier, mixed decision for highest confidence).
- **Training:** Conducted on NVIDIA RTX 5070 GPU with validation curves and early stopping.



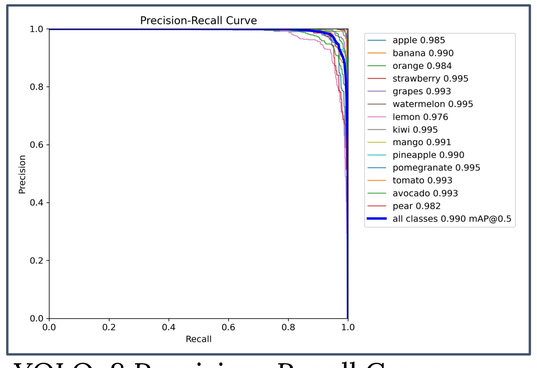
Detection Confusion Matrix



Classification Confusion Matrix



Validation Batch 1 Predictions



YOLOv8 Precision–Recall Curve

Results

- **Detection Performance (YOLOv8):** 97.99% precision, 95.34% recall, mAP@50 of 98.98%, and mAP@50–95 of 98.97%.
- **Classification Performance (MobileNetV2):** 95.99% precision, 95.61% recall, and 95.79% F1 score.
- **System Throughput:** ~5–6 FPS for detection and ~2 FPS for classification on Raspberry Pi 4B; >30 FPS combined on desktop PC.
- Most misclassifications occurred between visually similar fruits such as lemons and oranges.
- Maintained accuracy under varied lighting conditions and in cluttered, real-world image contexts.

Conclusions

Real-time fruit detection and classification can run entirely on a Raspberry Pi while maintaining near-desktop accuracy. The combined YOLOv8 and MobileNetV2 pipeline performed well under varied lighting and cluttered scenes.

Throughput on the Pi was lower but accuracy stayed consistent, showing that embedded systems can deliver practical computer vision performance when the pipeline is carefully designed.

Acknowledgments

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References

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